



ML PRODUCTION DEPLOYMENT

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ARTI 405



● Overview

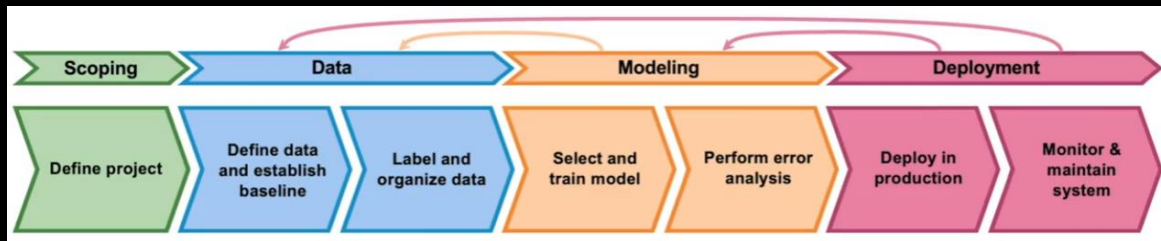
The goal of this assignment was to choose one of the provided Coursera courses, and relate its contents to the development environment being created in the Faraday room. I chose the course titled “The Machine Learning Project Lifecycle”, which gives a general run through of the process of creating and working on a machine learning project, with the goal of bringing it all the way to deployment.

This felt like a highly valuable course for me, as I have worked on quite a few ML projects, both personal and for school, but have never really gone through the process of bringing any of them to deployment, for use outside of my personal development environment. This is something that I have wanted to do for quite some time, but have found the information overload of the process to be rather daunting.



● Initial Key Takeaways

First and foremost, one of the most valuable things that I garnered from the course, which seems extremely simple, was this helpful visual diagram:



While the steps shown in this timeline seem rather obvious, I find great value in its simplicity. The entire process feels highly daunting, to the point that it is difficult to even begin researching. However, this simple diagram puts the process into such a tight, concise framing that really unscrambles my brain in relation to its concepts.

Later in the course, the instructor touches on typical questions that need to be asked during the actual deployment process, which relate to software engineering. For example, deciding whether requests need to be processed in real-time, or if they can be processed in batches. Another good example that was used is *where/how* the software is going to be run (e.g. in the cloud, at the browser).

These are questions that are asked in every software deployment, but thinking about them in the context of machine learning is a little bit more complex. This is because, generally speaking, machine learning tasks require quite a bit more processing than other forms of software, so it is important to consider where to allocate resources, and where compromises need to be made, in order for the deployment to remain user-friendly and satisfying.

● Deployment

This is the main part of the course that I found to be informational, as this is where my personal experience transitions into what I'm more unfamiliar with.

At the start of the section about the deployment process, there is mention of using bounding boxes, in order to compare human-input to the machine learning model's output. For a few years, I worked for a company, TELUS International, who performed this work on behalf of Google, so I actually have quite a bit of hands-on experience on the subject.

The purpose of doing this is to gather data of how the “learning algorithm” is performing, in relation to the direct input of a human being. This is, of course, prior to allowing the algorithm to perform real decisions.



The rough diagram above provides context to the process of human->AI transition that is outlined by the instructor. In an actual deployment, this sort of diagram would be applicable to some form of program that takes a specified user-input (e.g., images for object detection). From what I understood in the lesson, this type of framework would likely work best in a program that focuses on batch-processed requests.

● Relating to Faraday

So, the main goal of this assignment is to relate the course's information to the development environment that's being created in the Faraday room.

To start off, pretty much all of the course's content is related to the process of putting a project out into deployment, so any of the projects that would be utilizing the setup in the Faraday room would be in direct relation to the course. Its content isn't necessarily very specialized, and focuses mainly on providing a loose "guide" or reference of things to consider, as someone deploying a project.

Personally speaking, when visualizing the programs that I would like to use the development environment for, it's hard to see exactly how to go about executing each of the steps that are outlined in the course.

This is to say that, I can see the general steps, which were reinforced with diagrams, but I'm not sure *exactly* what these steps would specifically look like, in the context of a real project. I think that this is the main shortcoming of the course, as it doesn't reference/recommend any specific programs, or tools that should be used. I have a rough idea already (e.g., AWS buckets for bounding box), but I would have found it valuable to go through more specific examples.

